AI Infrastructure Resource Ledger: The Reality Check

Thursday, 5. March 2026

Engineer(s): Kali Hamilton

Project Title: Radar-VLA

Phase 1: The Calendar (Working Backward)

If you are still writing your training loop on your Drop-Dead Date, your project will fail.

1. Final Demo / Presentation Date: week of Apr 27, 2026 ??
2. Days required for Evaluation & Report: 14 days.
3. Estimated Training Run Duration: 30 days. (encoder pre-training + VLA tune)
4. DROP-DEAD TRAINING START DATE: ~Mar 14 *(Subtract Line 2 & 3 from Line 1)*

For ICRA April 3rd (S2S): drop-dead was already last week (fatal flaw)

Phase 2: The Stack & Compute Target

1. Base architecture backbone, include encoders etc., add as many lines as you need:
   - Radar encoder: Resnet-18 (11M params)
   - Vision: CLIP (86M)
   - LLM: Qwen2.5-1.5 B
     Param Count: ~1.6 Billion
2. Target Compute Hardware:
   [X] Local / Lab Node
   [ ] Cloud Rental (e.g., Lambda/AWS)
3. Specific GPU(s): 4090
   Total VRAM Available16 GB

Phase 3: The Memory Math (The VRAM Ledger)

Reference: 1B Parameters ≈ 2GB (Weights) + 2GB (Gradients) + 12GB (AdamW Optimizer States)

1. Model Weights (BF16): 1.5 x 2 = 3GB + 0.09B x 2 (CLIP frozen) = 3.18 B
2. Gradients (BF16):1.5x2 =3 GB
3. Optimizer States: (AdamW): **1.5Bx12 =18 GB**;
4. KV Cache & Activations Reserve: ~2-4?? GB
5. **TOTAL REQUIRED VRAM: 26 GB**

The Strategy: Is your Total Required VRAM > Total Available VRAM?

[] Yes [X] No (If I need to upgrade the model to 3B will need to steal Doncey's!!!)

If Yes, what is your exact mitigation strategy? [ ] FSDP / ZeRO-3 (Sharding across multiple GPUs)

[X] LoRA / QLoRA (Freezing base weights, avoiding optimizer state explosion)

trainable params only ~ 0.1 tops x 12 = 1.2 B

[ ] I am shrinking my model selection.

[ ] Something else (explain):

Phase 4: The Data Ingress (The I/O Bottleneck)

1. Dataset Size:16 TB / 58 driving sequences

2. Format:
   [X] Raw JPEGs/PNGs/trajectories/etc. in a folder (High I/O Risk);

   [ ] WebDataset / Sharded .tar
   [X] Something else: LIDAR point clouds, radar tesseracts

3. Storage Location:
   [X] Local NVMe SSD
   [ ] Network Drive
   [ ] S3/GCS Bucket
   [ ] Something else

If using raw images or other large data quantities over a network or standard SSD, how will you prevent your dataloader from starving your GPUs?

GAH. GREAT QUESTION. TBD

Phase 5: Red Team Audit (Partner Review)

Reviewing Engineer: Callie Jones

Look at your partner's ledger. Find the fatal flaw. Are they underestimating training time? Do they have a 40GB model fitting into 24GB of VRAM? Are they trying to load 2 million JPEGs sequentially?

The Fatal Flaw / Biggest Risk:

Data collection due to difficulties with having data that has radar geometry derived from lidar. Determining a possible pivot so a larger set of data can be used.

Phase 6: The 72-Hour Contract

What exact, granular engineering task will you complete in the next 72 hours to unblock these risks? (e.g., "Write dummy dataloader script to verify throughput," "Run a 10-step overfitting batch to check VRAM usage.")

Deliverable by Monday: picking a dataset strategy & scoping down project components

Phase 5: Red Team Audit (Partner Review)

Reviewing Engineer: Kali Hamilton

Look at your partner's ledger. Find the fatal flaw. Are they underestimating training time? Do they have a 40GB model fitting into 24GB of VRAM? Are they trying to load 2 million JPEGs sequentially?

The Fatal Flaw / Biggest Risk:

Finalizing architecture design & tools to be used such as simulating data.

Phase 6: The 72-Hour Contract

What exact, granular engineering task will you complete in the next 72 hours to unblock these risks? (e.g., "Write dummy dataloader script to verify throughput," "Run a 10-step overfitting batch to check VRAM usage.")

Deliverable by Monday:

Finalize architecture & tools and also an experiment training run at a small scale to confirm the setup is correct.